

Lab1(answers)

Task 1.1

Relation A: Employee:

1. Six superkeys: {EmpID}, {SSN}, {Email}, {Phone}, {EmpID, Name}, {SSN, Department}
2. Candidate keys: {EmpID}, {SSN}, {Email}, {Phone}
3. Primary key: {EmpID} (since email and phone can be changed and SSN is sensitive data)
4. The data shows that all three employees have different phone numbers, but this alone does not guarantee that phone numbers must be unique. Unless it's given in Business Rules:

Relation B: Course Registration:

1. Minimum attributes for the Primary Key :
(StudentID, CourseCode, Section, Semester, Year)
2. Explanation :
StudentID (different students can sign up for the same course)
CourseCode (single student can take different courses in the same semester)
Section (single course can have multiple sections)
Semester (student can take the same course in different semesters)
Year (to determine specific academic period)
3. No, based on the Business Rules.

Task 1.2: Foreign Key Design:

1. Student.AdvisorID → Professor.ProfID
2. Course.DepartmentCode → Department.DeptCode
3. Department.ChairID → Professor.ProfID
4. Enrollment.StudentID → Student.StudentID
5. Enrollment.CourseID → Course.CourseID

Part 4: Normalization Workshop

Task 4.1

1. Functional Dependencies:

StudentID → StudentName, StudentMajor

ProjectID → ProjectTitle, ProjectType, SupervisorID

SupervisorID → SupervisorName, SupervisorDept

(StudentID, ProjectID) → Role, HoursWorked, StartDate, EndDate

2. Problems:

Student information is repeated for every project.

Project information is repeated for every student.

Supervisor information is repeated for every project/student.

Update anomaly: Changing a student's major or supervisor's department requires updating multiple rows.

Insert anomaly: A new project cannot be inserted if no student is assigned to it.

Delete anomaly: Deleting the only student assigned to a project may also delete all information about that project.

3. 1NF:

No 1NF violations. All attributes contain atomic values.

2NF Decomposition:

Student(StudentID PK, StudentName, StudentMajor)

Project(ProjectID PK, ProjectTitle, ProjectType, SupervisorID)

StudentProject(StudentID PK/FK, ProjectID PK/FK, Role, HoursWorked, StartDate, EndDate)

3NF

Transitive Dependencies:

ProjectID → SupervisorID

SupervisorID → SupervisorName, SupervisorDept

Therefore:

ProjectID → SupervisorName, SupervisorDept

Final 3NF Decomposition:

Student(StudentID PK, StudentName, StudentMajor)

Supervisor(SupervisorID PK, SupervisorName, SupervisorDept)

Project(ProjectID PK, ProjectTitle, ProjectType, SupervisorID FK)

StudentProject(StudentID PK/FK, ProjectID PK/FK, Role, HoursWorked, StartDate, EndDate)

Task 4.2 — BCNF Decomposition

1. Primary Key:

(StudentID, CourseID, TimeSlot, Room)

2. Functional Dependencies:

StudentID → StudentMajor

CourseID → CourseName

InstructorID → InstructorName

(TimeSlot, Room) → Building

(CourseID, TimeSlot, Room) → InstructorID

3. BCNF:

The table is not in BCNF because some determinants are not superkeys.

Violations:

StudentID $\rightarrow$ StudentMajor

CourseID $\rightarrow$ CourseName

InstructorID $\rightarrow$ InstructorName

(TimeSlot, Room) $\rightarrow$ Building

(CourseID, TimeSlot, Room) $\rightarrow$ InstructorID

4. BCNF Decomposition:

Student(StudentID PK, StudentMajor)

Course(CourseID PK, CourseName)

Instructor(InstructorID PK, InstructorName)

RoomSchedule((TimeSlot, Room) PK, Building)

CourseSection(CourseID PK/FK, TimeSlot PK, Room PK, InstructorID FK)

Enrollment(StudentID PK/FK, CourseID PK/FK, TimeSlot PK/FK, Room PK/FK)

5. Loss of Information:

The decomposition is lossless. The original information can be reconstructed by joining the decomposed tables using their primary and foreign keys.

Part 5

2)

STUDENT(StudentID PK, Name, Email, Major)

FACULTY(FacultyID PK, Name, Email, Department)

CLUB(ClubID PK, ClubName, Description, FoundingDate, FacultyID FK)

MEMBERSHIP(StudentID PK/FK, ClubID PK/FK, JoinDate, MembershipStatus)

EVENT(EventID PK, EventName, EventDate, StartTime, EndTime, Description, ClubID FK)

ATTENDANCE(StudentID PK/FK, EventID PK/FK, Attended)

OFFICER_POSITION(PositionID PK, PositionName)

CLUBOFFICER(StudentID PK/FK, ClubID PK/FK, PositionID PK/FK,StartDate, EndDate)

BUDGET(BudgetID PK, ClubID FK, Amount, Year)

EXPENSE(ExpenseID PK, ClubID FK, Amount, ExpenseDate, Description)

ROOM(RoomID PK, Building, RoomNumber, Capacity)

ROOM_RESERVATION(ReservationID PK, EventID FK, RoomID FK,ReservationDate)

3)

“Find all students who are members of the Computer Science Club.”

“List all upcoming club events with their reserved rooms, building names, and room capacities.”

“Calculate the total expenses and remaining budget for each student club.”